

# Phase-Change Memory: The Likely Successor to Flash Memory

Technology analysts forecast that in the next two to four years, phase-change memory (PCM) chips and modules will drastically change the memory landscape, enabling lower-cost but higher-performance digital products of the future. This article discusses PCM technology in detail while also covering its comparison with other memory technologies and future outlook

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It is much easier to find our way across a neat and well-ordered store than a messy and disorganised one. Electronic conduction shows a similar preference; ordered, crystalline materials make much better conductors than disordered, amorphous materials. Single-crystal conductors show much lower resistance to electric current than their polycrystalline or amorphous forms. This is the reason why almost all electronic devices are made from monocrystalline semiconductors.

The difference in resistivity between crystalline and amorphous forms of the same conductor can be very big. This should not come as a surprise because, after all, resistivity is a physical property that can vary over an enormous range with the ratio of resistivity between good and bad conductors easily exceeding  $10^{18}$ , at a given temperature.

The remarkable contrast between the resistivities of ordered and disordered phases of certain conducting materials is the basis of an emerging memory technology called phase-change memory (PCM), which promises to revolutionise the digital system design in the near future. It has taken many years and sustained efforts of many individuals and companies for phase-change technology to leave the confines of research laboratories and become a commercial reality. By all indications, this is a revolutionary new device with much promise for future applications.

|                  | Dynamic Random Access Memory (DRAM) | Flash             | Phase-Change Memory (PCM)                                                   | Multilevel Phase-Change Memory |
|------------------|-------------------------------------|-------------------|-----------------------------------------------------------------------------|--------------------------------|
| <b>Timeframe</b> | Invented in 1966                    | Invented in 1980s | Available in limited number of smartphones - wide adoption expected by 2016 | 2016                           |
| <b>Speed</b>     | Green                               | Red               | Yellow                                                                      | Yellow                         |
| <b>Density</b>   | Yellow                              | Green             | Green                                                                       | Green                          |
| <b>Endurance</b> | Green                               | Red               | Yellow                                                                      | Yellow                         |
| <b>Retention</b> | Red                                 | Red               | Green                                                                       | Green                          |
| <b>Scaling</b>   | Orange                              | Yellow            | Green                                                                       | Green                          |

Best in class

Average/Inadequate

Good/Adequate

Bad/Worse in class

Fig. 1: PCMs Vs DRAMs and Flash memories

## The science of PCM material

Elements belonging to group VI of the periodic table are called chalcogens. These elements can combine with other electropositive elements to form semi-metallic compounds called 'chalcogenides.' Some of these compounds have a remarkable property of being able to exist in either a crystalline or an amorphous state with very small transition energy to go from one state to the other.

If such a material is melted and then cooled quickly, its constituent atoms do not get the time to settle in an ordered crystal lattice, producing a disordered metallic glass phase having high resistivity. However, if the molten material is cooled more slowly, a low-resistivity crystalline phase forms.

What is astounding is that the tran-

sition from one phase to another can be affected in just a few tens of nanoseconds (ns). The high phase-transition speed and the huge difference in resistivity between the crystalline and amorphous phases in a phase-change chalcogenide can be used to store the binary data.

Phase-change memories use a material called 'germanium antimony telluride' ( $\text{Ge}_2\text{Sb}_2\text{Te}_3$ ), abbreviated as GST. Its low-resistivity crystalline phase can be used to represent binary '1' (set state), whereas its high-resistivity amorphous phase can be used to code for binary '0' (reset state). One can change the logic state by controlled heating of the phase-change material—rapidly heating and cooling the molten material quenches the material to amorphous phase, whereas somewhat slower heating anneals the material to its crystalline state. In a memory